AIRA

APNEA & INFANT RESPIRATION ANALYSIS

Contactless, explainable apnea alerting from a camera: how it works, the problems faced, and how each was solved.

Diuthendrh R D, 2026. Research prototype, not a medical device.

1. The idea

Newborns, especially preterm babies, can suddenly stop breathing (apnea). Hospitals watch for this with chest electrodes and pulse oximeters stuck to very fragile skin. AIRA does it with only a camera: it watches the chest, counts every breath in real time and alerts staff when breathing stops. One camera can watch several cots.

Core principle: every stage is classical, explainable signal processing that can be traced to a known equation, rather than a black-box model. And **uncertainty must fail toward caution, never toward silence**: if AIRA cannot see well, it says so.

2. How it works (pipeline)

Stage	What it does	Technique
1. Find the breathing area	Scans the picture for anything that rises and falls at a breathing rhythm, and locks onto the chest or belly. No clicking.	Block-wise spectral analysis (rhythm concentration and coherence), MediaPipe body landmarks
2. Measure motion	Measures tiny frame-to-frame movement in a 4 x 4 grid (32 motion signals, every direction).	Farneback dense optical flow
3. Fuse the signals	Weights each of the 32 signals by how much real breathing it carries compared with noise, into one breathing wave.	Max-SNR channel fusion (generalized eigenvector)
4. Clean the signal	Keeps only breathing-speed motion.	Causal Butterworth band-pass filter
5. Count breaths	Detects each breath with thresholds learned from this baby, above the measured noise floor.	Adaptive hysteresis detector
6. Apnea clock	Time since the last real breath: watch at 10 s, warning at 15 s, alarm at 20 s. The alarm stays on until a nurse confirms it.	Tiered, latched alert
7. Heart rate (beta)	Reads the pulse from tiny skin colour changes on the face.	rPPG: POS and CHROM methods, best one kept

3. Problems faced and how they were solved

Finding the baby (ROI detection)

> **Problem:** Pose models are trained on upright adults and often fail on swaddled, side-lying or blanket-covered babies, so a pose-only chest box was unreliable.
> **Solution:** Find the breathing area from the breathing itself: AIRA looks for regions that move at a steady breathing rhythm. Body landmarks are used when a person is visible, and the last good area is kept if they are briefly lost.

> **Problem:** Webcam sensor noise and light flicker also "breathe" at a rhythm, so the camera's own noise was detected as a baby.
> **Solution:** Reject rhythms that cover most of the picture or its edges (camera shimmer), remove isolated noisy blocks, and rank candidate areas by strength and coherence.

> **Problem:** Finding the area and learning the breathing took too long.
> **Solution:** Shorter scan and learning windows, tested on held-out babies to make sure accuracy and false alarms did not get worse.

Movement

> **Problem:** A kick, a squirm or a caregiver touching the blanket looks like a huge breath.
> **Solution:** Sudden large, non-rhythmic motion is detected (well above the recent normal level) and pauses breath counting instead of faking breaths. The apnea clock keeps running, so movement can never hide an apnea.

> **Problem:** If the breathing area keeps moving, the signal breaks; if it never moves, a baby who rolls over is lost.
> **Solution:** The area stays still while the baby is still, and moves only after a real, sustained shift of the body.

Side-lying babies

> **Problem:** Measuring only up-and-down motion assumes the baby lies on their back. A side-lying baby's chest moves sideways.
> **Solution:** Motion is measured in every direction, and the fusion step finds whichever direction carries the breathing, for any position.

Every baby is different

> **Problem:** One fixed threshold for all babies either misses shallow breathers or false-alarms on others.
> **Solution:** AIRA first learns each baby's own normal breathing (depth and rate) and keeps adapting slowly. Alarms work from the first second, using a cautious default until learning is done.

> **Problem:** "Boiling frog": a slowly adapting baseline could drift and quietly accept worsening breathing as normal.
> **Solution:** Only clean breaths update the baseline, and big changes are not absorbed silently.

Bad view and blind spots

> **Problem:** A covered baby, a dark room or a poor picture can look like "no breathing" or, worse, like normal breathing.
> **Solution:** A signal-quality check raises a "poor signal / check camera" technical alert. AIRA never silently assumes a baby is fine.

> **Problem:** A baby taken out and another put in the same cot must not inherit the previous baby's learned breathing.
> **Solution:** Staff can mark a cot empty or occupied, and a new baby always triggers fresh learning. When in doubt, re-learn: it is the safe default.

Camera glitches

> **Problem:** The webcam repeated identical frames, which made a breathing person look perfectly still and broke the movement checks.
> **Solution:** Duplicate frames are detected and skipped, and the frame rate is measured from real new frames only.

Heart rate (beta)

> **Problem:** The pulse reading jumped around and dropped in and out.
> **Solution:** Track the face with Face Mesh, run two pulse methods and keep the cleaner one, ignore single spikes, and hold the last steady value through short dropouts. It still needs good light, a visible face and a colour camera, so it stays labelled beta.

4. Results

Tested on the public **AIR-125** dataset (real infant videos with every breath marked by annotators). Settings were tuned on 2 babies and tested once on 3 babies the system had never seen (63 clips):

Measure	Result
Breaths found	88%
Detections that were real breaths	90%
Breathing rate error	About 1 breath per minute on average (e.g. experts count 40/min, AIRA says 39-41)
Clips with a false apnea alarm	0 of 63

Apnea alarms were tested on simulated babies with real 20-40 s apneas, to check that alarms come on time and that almost no breaths are reported while a baby is not breathing. To improve breath detection, 12 different approaches were compared under real-time limits; the best (max-SNR fusion) is what AIRA uses.

5. AIRA vs. real-life apnea monitoring

	Hospital monitor (today)	AIRA (camera)
Contact	Electrodes and sensors on the skin	None
Breathing measured by	Chest impedance (electrodes)	Chest motion (optical flow)
Central apnea (no breathing effort)	Detected	Detected
Obstructive apnea (airway blocked, chest still moves)	Missed by breathing channel; caught later by oxygen/heart-rate drop	Missed today; thermal camera planned
Status	Approved medical devices	Research prototype

Obstructive apnea is the hard case: the airway is blocked, so the baby keeps trying to breathe and the chest keeps moving, but no air flows. Any motion-based monitor, including hospital electrodes, still sees "breathing". A **thermal camera** can see the warm air leaving the nose; no airflow means no breath, even when the chest moves. Combining chest motion (camera) with airflow (thermal) would catch both kinds.

6. Limits and future scope

AIRA was built and tested on a **laptop and its webcam**. That limited hardware is its biggest constraint:

- **Infrared (night-vision) camera:** monitoring in complete darkness, where babies actually sleep.
- **Thermal camera:** airflow at the nose, to catch obstructive apnea.
- **Second sensor, an under-mattress pad:** a piezo film or load cell under each cot feels breathing through the bedding, so monitoring continues under blankets, in darkness or with the baby turned away. Camera and pad are fused: an apnea alarm when neither detects breathing, a lower-confidence "check baby" warning when only one loses the signal. One pad per cot also gives every baby their own signal. Like the camera it senses movement, so obstructive apnea still needs the thermal camera.
- **Multi-baby detection and tracking** (designed in the original methodology with YOLO and ByteTrack) for crowded wards.
- **Clinical validation** side by side with wired NICU monitors, and alarms sent to the nurse station.

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AIRA is a research prototype and is not approved for clinical use. Dataset: AIR-125 (Infant Respiration Estimation, Ostadabbas lab). Body and face landmarks: Google MediaPipe. rPPG: POS (Wang et al., 2017) and CHROM (de Haan and Jeanne, 2013).